ApplyCanary — architecture

An agentic job-search assistant. CockroachDB is the memory layer; AWS runs the models and the voice.

Browser

React dashboard

Jobs, Match cards, Interview Studio, Memory

`/api` · guest mode

Serves SPA + API from Render

HTTP /api

FastAPI agent

Amazon ECS · Fargate

- Web API
- Scheduler
- LLM chain + circuit breakers
- Role discovery
- Truthcheck gate
- Interview state machine

Claude · Titan · TTS · STT

AWS

- Amazon Bedrock — Claude + Titan embeddings
- Amazon Polly — spoken interviewer
- Amazon Transcribe — hears answers
- Amazon S3 — interview audio
- CloudWatch — logs & metrics

postgres wire (transactions · vectors)

CockroachDB Serverless — the memory layer

Transactional state — jobs · scores · sessions · turns · applications

Distributed vector indexing — native VECTOR(1024) columns on job_embedding & agent_memory, backed by a vec_cosine_ops index

Semantic recall is SQL — 1 − vec_cosine_distance(embedding, :q::vector)

- MCP Server — read-only, audited agent access
- ccloud CLI — provision · backups · audit
- Agent Skills — cockroachlabs/cockroachdb-skills

read-only · audited